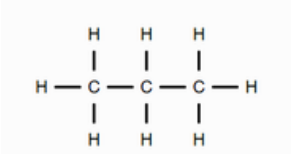
ONE MOLECULE, FOUR REPRESENTATIONS:
From Molecular Formula to Bond-Line Structure

CHEM 13 - Organic Chemistry (Lecture) | BS Chemical Engineering

Katrina Faye A. Soncuya
2B-BSChE

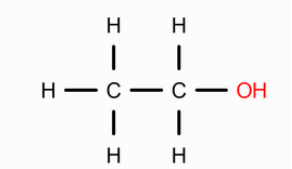
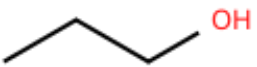
Propane

Molecular Formula: C_3H_8

C_3H_8 Molecular Formula:	 Lewis Structure:
$CH_3CH_2CH_3$ Condensed Structural Formula:	 Bond-Line Formula:

Ethanol

Molecular Formula: C_2H_6O

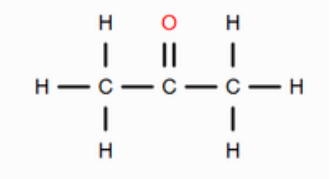
C_2H_6O Molecular Formula:	 Lewis Structure:
CH_3CH_2OH Condensed Structural Formula:	 Bond-Line Formula:

Acetone (Propan-2-one)

Molecular Formula: C_3H_6O

C_3H_6O

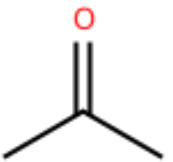
Molecular Formula:



Lewis Structure:

$CH_3C(=O)CH_3$

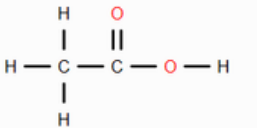
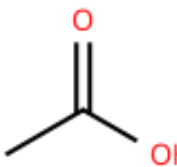
Condensed Structural Formula:



Bond-Line Formula:

Acetic acid (ethanoic acid)

Molecular Formula: $C_2H_4O_2$

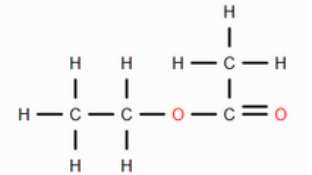
$C_2H_4O_2$ Molecular Formula:	 Lewis Structure:
CH_3COOH Condensed Structural Formula:	 Bond-Line Formula:

Ethyl Acetate (ethyl ethanoate)

Molecular Formula: $C_4H_8O_2$

$C_4H_8O_2$

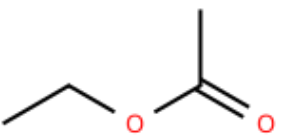
Molecular Formula:



Lewis Structure:

$CH_3COOH_2CH_3$

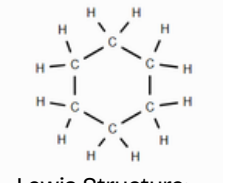
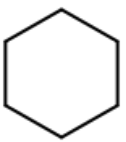
Condensed Structural Formula:



Bond-Line Formula:

Cyclohexane

Molecular Formula: C_6H_{12}

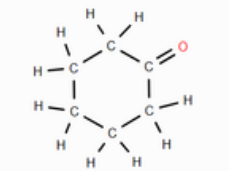
C_6H_{12} Molecular Formula:	 Lewis Structure:
$CH_2CH_2CH_2CH_2CH_2CH_2$ Condensed Structural Formula:	 Bond-Line Formula:

Cyclohexanone

Molecular Formula: $C_6H_{10}O$

$C_6H_{10}O$

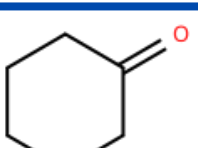
Molecular Formula:



Lewis Structure:

$(CH_2)_5CO$

Condensed Structural Formula:



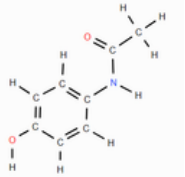
Bond-Line Formula:

Acetaminophen (paracetamol)

Molecular Formula: $C_8H_9NO_2$

$C_8H_9NO_2$

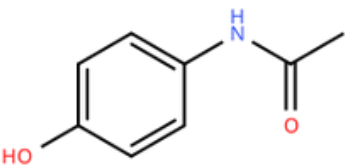
Molecular Formula:



Lewis Structure:

$HOC_6H_4NHCOCH_3$

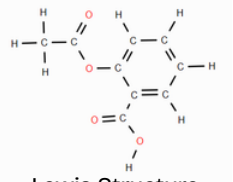
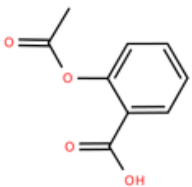
Condensed Structural Formula:



Bond-Line Formula:

Aspirin (acetylsalicylic acid)

Molecular Formula: $C_9H_8O_4$

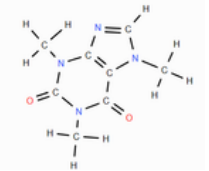
$C_9H_8O_4$ Molecular Formula:	 Lewis Structure:
$CH_3COOC_6H_4COOH$ Condensed Structural Formula:	 Bond-Line Formula:

Caffeine

Molecular Formula: $C_8H_{10}N_4O_2$

$C_8H_{10}N_4O_2$

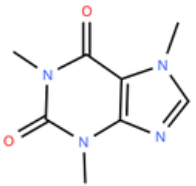
Molecular Formula:



Lewis Structure:

$CH_3N_4C_5O_2(CH_3)_2$

Condensed Structural Formula:



Bond-Line Formula: